

Automotive Ethernet Fuzzing: From Purchasing ECU to SOME/IP Fuzzing

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AUTOCRYPT

Who are we?



a Mobility Security Company
for Autonomous, Connected Vehicles, and Beyond

- Jonghyuk Song (jhsong@autocrypt.io)
 - AUTOCRYPT Red Team leader, automotive cyber security researcher
 - Offensive security researcher, CTF player (Defcon CTF Finalist 2009, 2011, 2017)
- Soohwan Oh
 - Automotive engineer and security tester at AUTOCRYPT
 - Experts for fuzzing test and issue analysis on the in-vehicle networks
- Woongjo Choi
 - AUTOCRYPT Blue Team leader, automotive security tester
 - Software test expert : vehicle, mobile, application processor, ultrasound system, etc.

This talk is for

- How to setup test environment with real ECUs?
 - How to do Automotive Ethernet fuzzing on the ECU?
- A practical guide to SOME/IP fuzzing with the real ECUs.

Advent of Automotive Ethernet

Higher data bandwidth is required for ADAS, infotainment, ...

CAN is not enough to handle it.

Automakers have started to adopt Automotive Ethernet.

BMW X3, X4, X5, i3, i8, ...

Jaguar XJ, XF, ...

Volkswagen Passat

In-Vehicle Infotainment



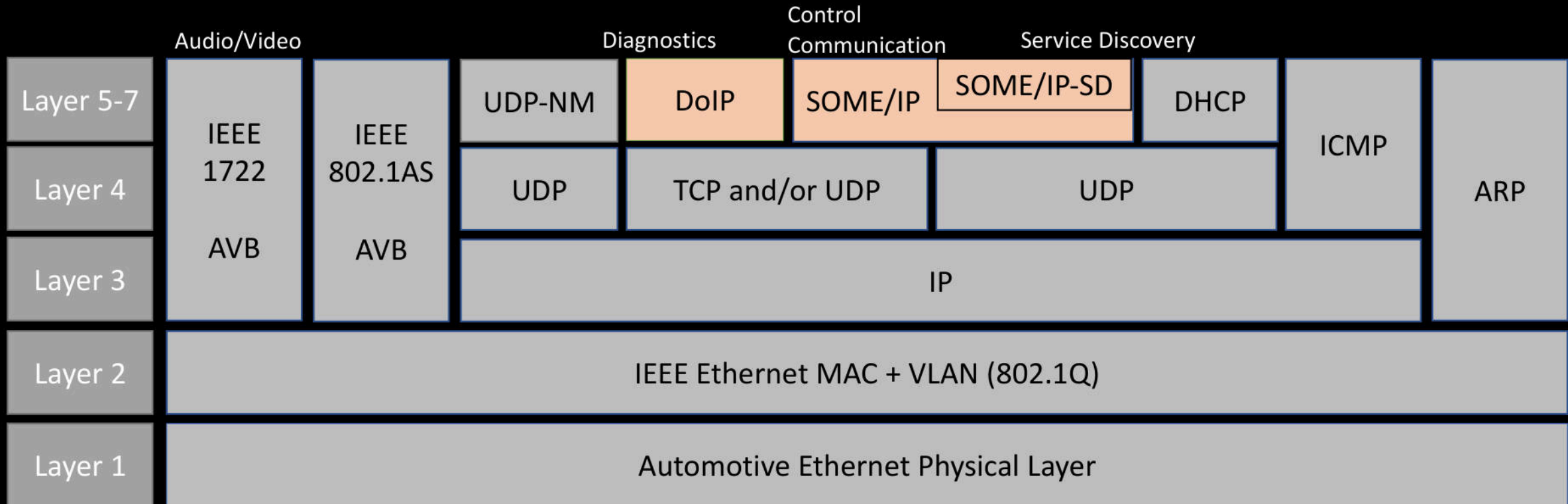
Diagnostics



Camera System for ADAS

(Advanced Driver Assistance System)

Automotive Ethernet Protocols



- SOME/IP (Scalable service-Oriented MiddlewarE over IP)
 - Control communication protocol between ECUs. (RPC, event notifications)
- DoIP (Diagnostics Over IP)
 - Automotive diagnostic protocol based on IP

Automotive Ethernet Fuzzing?

Automotive Ethernet has not yet been tested enough.

→ Let's do SOME/IP fuzzing!

→ Is it essential buying a car?

Let's test on ECU

It's expensive to buy a whole car for every single test.

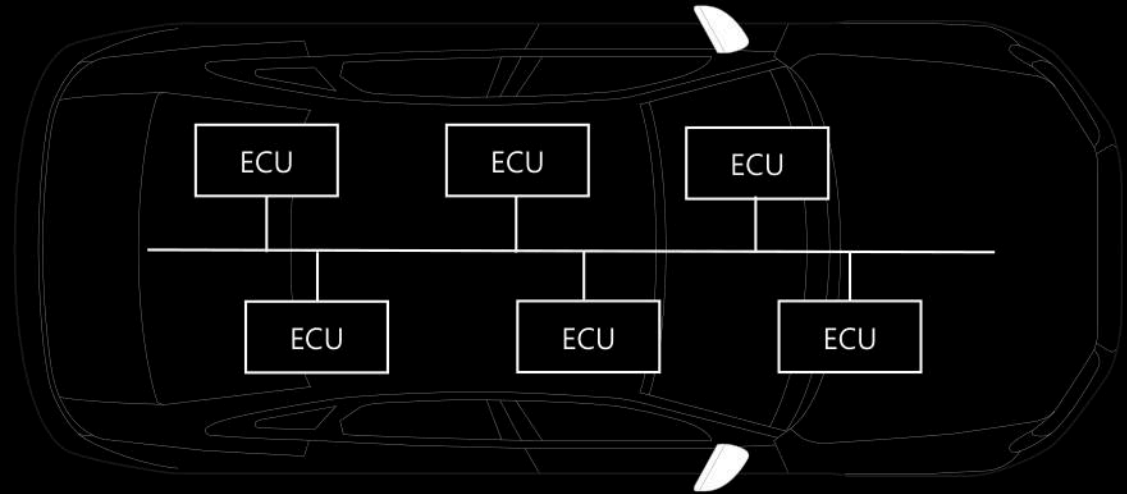
→ Let's hack an ECU first!

How?

Where can we buy?

Which ECU is proper for test?

Then how to setup?



Procedures

1. Select ECU
2. Buy ECU
3. Setup test environment
4. Network configuration for SOME/IP
5. Discovery SOME/IP services
6. Fuzzing

Select ECU

ECU Survey & Selection

- We chose a BMW iX head unit ECU.
 - Most head unit ECU support Automotive Ethernet.
- Why BMW?
 - Automotive Ethernet industry leader
 - First automaker to use Ethernet in a production vehicle.
(X5 model uses Ethernet in camera system at 2013)
 - BMW provides ECU info on TIS (Technical Information System)
 - wiring diagrams, pin map, connector, basic info, ...

BMW Technical Information System

Technical Information System

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Registration and payment are required to enter this site. Already registered? Provide your login information to the left and select the submit button. Unregistered users please signup and you will be redirected to select one of the plans below on login.

\$30 1 Day (24 hour) Subscription

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\$2500 1 Year Subscription

Paid service

Search for vehicle

with vehicle identification number

WB5 [redacted]

With basic features

Brand

With type code

Start new search

VIN number is required to search

Wiring diagrams Search within results

Service Information B... Flat Rate Units Packages Defects Repair overviews (part... Parts Repa

Technical data NED Repair and mainten... Operating fluids SIT Technology Wiring diagrams

385 Wiring diagrams

+ Component structure

+ Function structure

Expanded vehicle cha... ☐

A11 Cras
Wiring di
Technical - V.3

A218 Tai
Wiring di
Technical - V.1

A21 Roo
Wiring di
Technical - V.3

A255 Tra
Wiring di
Technical - V.2

A26 Fron
Wiring di
Technical - V.2

A343 Ult
Wiring di
Technical document - V.1

A402 Headunit High (HU-H)
Wiring diagram I SSP-BTS-SP0000068755 - V.5
Technical document

Censored

Google serves various VIN numbers

A screenshot of a Google search results page for the query "bmw ix vin number". The search bar at the top shows the query and standard navigation icons. The first result is from dealerrater.com, titled "2022 BMW iX, new, \$97570 VIN WB533CF00N0K51701". The VIN is highlighted with a red box. The snippet mentions the car is priced at \$97570, has 0 miles, and is Storm Bay Metallic. It includes a 3.7-star rating and 719 reviews. The second result is from youcanic.com, titled "Where is the VIN number on a BMW? - YOUCANIC". The snippet explains that the VIN can be found in several locations on the vehicle. The third result is from vehicleidentificationnumber.com, titled "BMW - Vehicle Identification Number". The snippet states that the VIN can be found at various locations on the car. The fourth result is from group1auto.com, titled "New 2023 BMW iX M60 VIN: WB533CF00N0K51701". The VIN is again highlighted with a red box. On the right side of the search results, there are two images: a blue BMW iX car and a close-up of a car's interior showing the VIN plate. A small "Google 홈페이지로" (Return to Google homepage) button is visible near the bottom right image.

2022 BMW iX

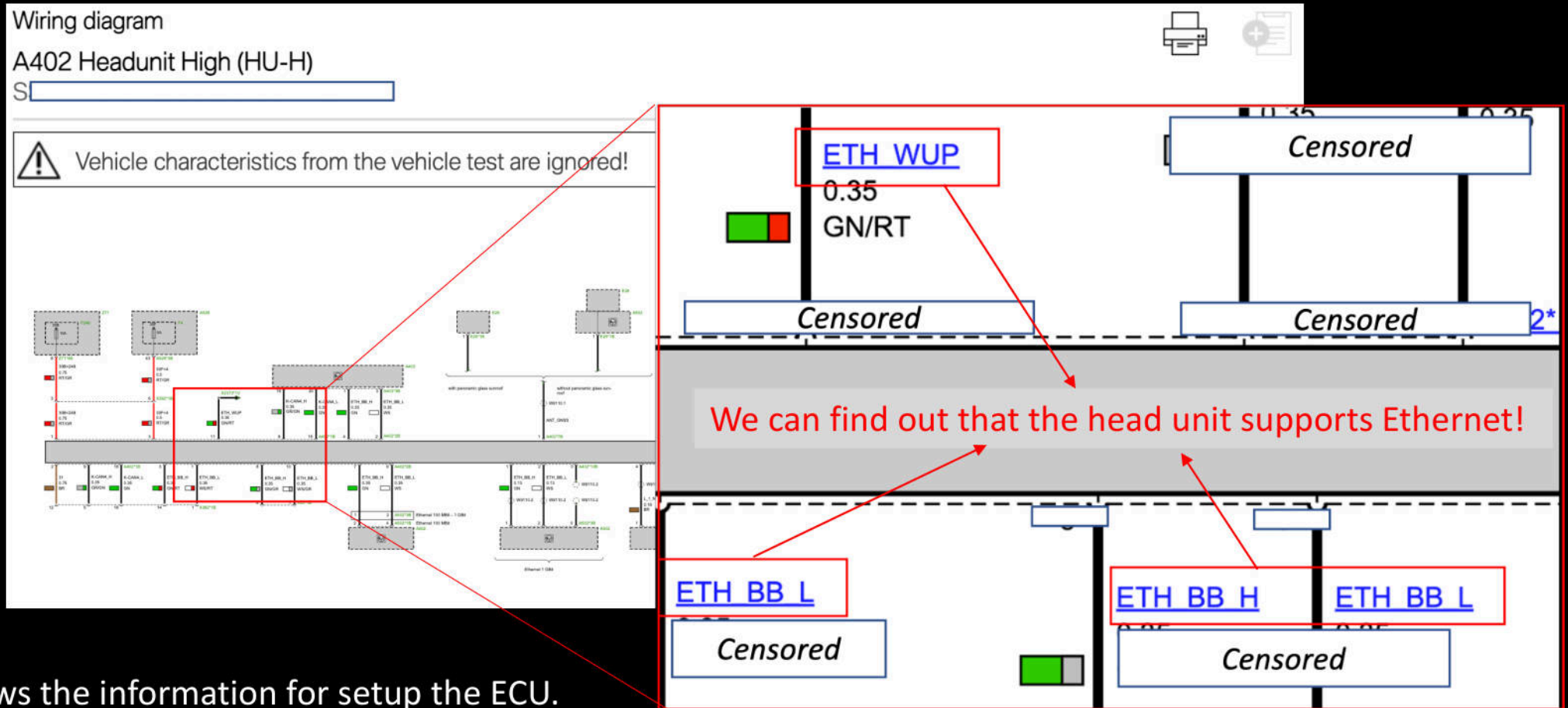
\$91,245 • new



Basics

Condition	New
Interior Color	Black
Exterior Color	Sophisto
Drivetrain	All-Wheel Drive
Transmission	Automatic
Fuel	Electric
Engine	Electric
VIN	WB53CFC02NCH000000
Stock Number	CK57882

Wiring Diagram



Wiring diagram shows the information for setup the ECU.
power, ground, ETH, ...

Buy ECU

Gain ECUs from real automakers

- In South Korea,
 - BMW
 - Some service centers sell ECUs.
 - We searched the BMW service centers from BMW Korea homepage → Phone call → Buy
 - Tesla
 - Service centers sell ECUs but not as an individual unit.
 - The ECUs should be installed in the car.
 - Hyundai
 - Hyundai Mobis agency sells ECUs.

Used ECUs are able to obtain from eBay

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Shop by category All Categories

Related: [bmw headunit](#) ☐ Include description

Category: **All** eBay Motors Car & Truck Parts & Accessories Other Car & Truck Parts & Accessories Other Car & Truck Interior Parts & Accessories Car Stereos & Head Units Car & Truck Light Bulbs & LEDs More

Brand: ☐ ALPINE (74) ☐ Becker (25) ☐ Blaupunkt (1) ☐ BMW (1,393) ☐ Eonon (113) ☐ HIZPO (60) ☐ Unbranded (272) ☐ Xtrons (37) [See all](#)

Condition: ☐ New (20,332) ☐ Used (3,549) ☐ Not Specified (2) [See all](#)

Price: ☐ Under \$35.00 ☐ \$35.00 to \$350.00 ☐ Over \$350.00 \$ Min to \$ Max

Available inventory

22,000+ results for **bmw head unit** [Save this search](#) Shipping to: Korea, South

Category: Car Stereos & Head Units Other Car & Truck Interior Parts & Accessories Built-in Car Sat Nav Devices

Android 10.0 OS
4-Core 2GB RAM+32GB ROM NBT CarPlay

Android 9.1
For BMW 3 Series E46 M3 Power 71 MG 27

Android 9.1
For BMW E38 E39 E53 X5 GPS Navigation DAB Android 9.1 Car...

BMW F30 NBT Entry Head Unit Repair Service F30/06/32/34
Includes: It's called to your car after repair. [Contact us for updates!](#)

Android 10.0 Head Unit Stereo Radio GPS SAT Navi For BMW 5...
Brand New
\$393.19 Was: \$443.98 5% off or Best Offer
Free International Shipping from China
Last one Sponsored

7" Android 10 Head Unit Radio GPS SAT Navi For BMW 3 Series...
Brand New
\$177.04 Was: \$188.36 6% off or Best Offer
Free International Shipping from China
Last one Sponsored

For BMW E38 E39 E53 X5 GPS Navigation DAB Android 9.1 Car...
Brand New
\$152.48 or Best Offer
+\$27.62 shipping estimate from United States
Free returns
25 sold Sponsored

BMW F30 NBT Entry Head Unit Repair Service 9290998...
Refurbished
\$450.00 Buy It Now
+\$50.00 shipping from United States
47 sold Sponsored

For BMW X3 E83 2003-2010 Car Stereo Radio DAB+Android 10...
Brand New
\$212.33 or Best Offer
+\$14.40 shipping from China
Last one

7" Android 10 Head Unit Radio GPS SAT Navi For BMW 3 Series...
Brand New
\$202.74 or Best Offer
Free International Shipping from China
Last one

For BMW 325i 328i 330i 335i 9 INCH Android Radio GPS...
Brand New
\$210.00 or Best Offer
Free International Shipping from China

XTRONS® 11.6" Android 11 Car GPS Stereo 8-Co...
New (Other)
\$368.99 Buy It Now
+\$25.00 shipping from Hong Kong
5 watchers

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Shop by category All Categories

Category: **All** eBay Motors Car & Truck Individual Gauges Other Car & Truck Interior Parts & Accessories Car & Truck Body Moldings & Trims Car & Truck ECUs & Computer Modules In-Car Head Up Displays (HUDs) More

Brand: ☐ Tesla (11) ☐ Unbranded (60) ☐ Not Specified (15)

Brand Type: ☐ Genuine OEM (11) ☐ Unbranded (60) ☐ Not Specified (10)

Color: ☐ Material: ☐ Manufacturer Warranty: ☐ Type: ☐ Condition: ☐ New (112) ☐ Used (16) [See all](#)

Price: ☐ Under \$35.00

130 results for **tesla car computer** [Save this search](#) Shipping to: Korea, South

Find the right parts for your Tesla

TESLA X S GPS Driver Assist Autopilot CAR COMPUTER ECU...
Pre-Owned
\$509.60 or Best Offer
+\$91.91 shipping from Poland
Sponsored

2017 - 2021 TESLA MODEL 3 CAR MEDIA COMPUTER CONTROL...
Pre-Owned
\$899.93 or Best Offer
+\$43.09 shipping estimate from United States
Benefits charity Sponsored

TESLA 3 S GPS NAVIGATION Driver Assist Autopilot CAR...
Pre-Owned
\$509.60 or Best Offer
+\$91.91 shipping from Poland
Sponsored

Car HUD Overspeed Warning Trip Computer T20-HUD For Tesla...
Brand New
\$69.85 Was: \$77.64 10% off or Best Offer
Free International Shipping from China
Sponsored

2017 - 2019 TESLA MODEL 3 CAR COMPUTER MEDIA CONTROLLE...
Pre-Owned
\$188.00 or Best Offer
+\$58.35 shipping estimate from United States

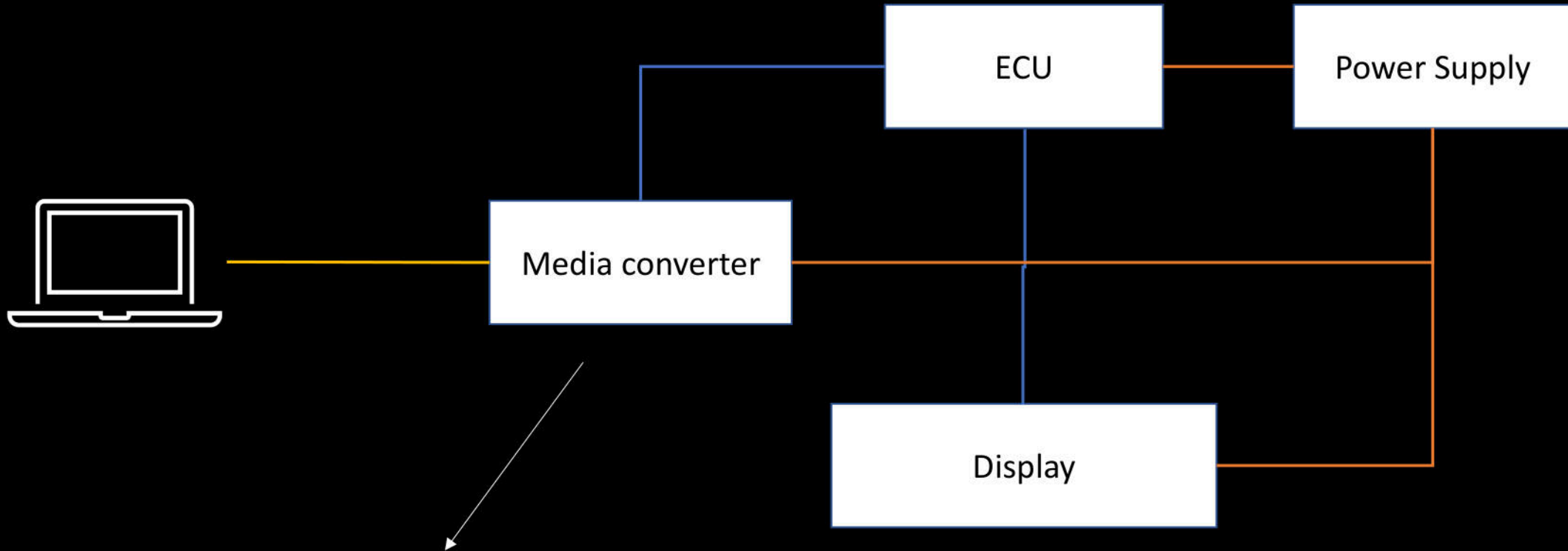
2017-2021 TESLA MODEL 3 CAR MEDIA COMPUTER CONTROL...
Pre-Owned
\$695.00 or Best Offer
Shipping not specified from United States

Tesla Model 3 Car Computer MCU Module Mount Bracket
Pre-Owned
\$50.55 Was: \$64.96 8% off or Best Offer
+\$22.74 shipping estimate from United States

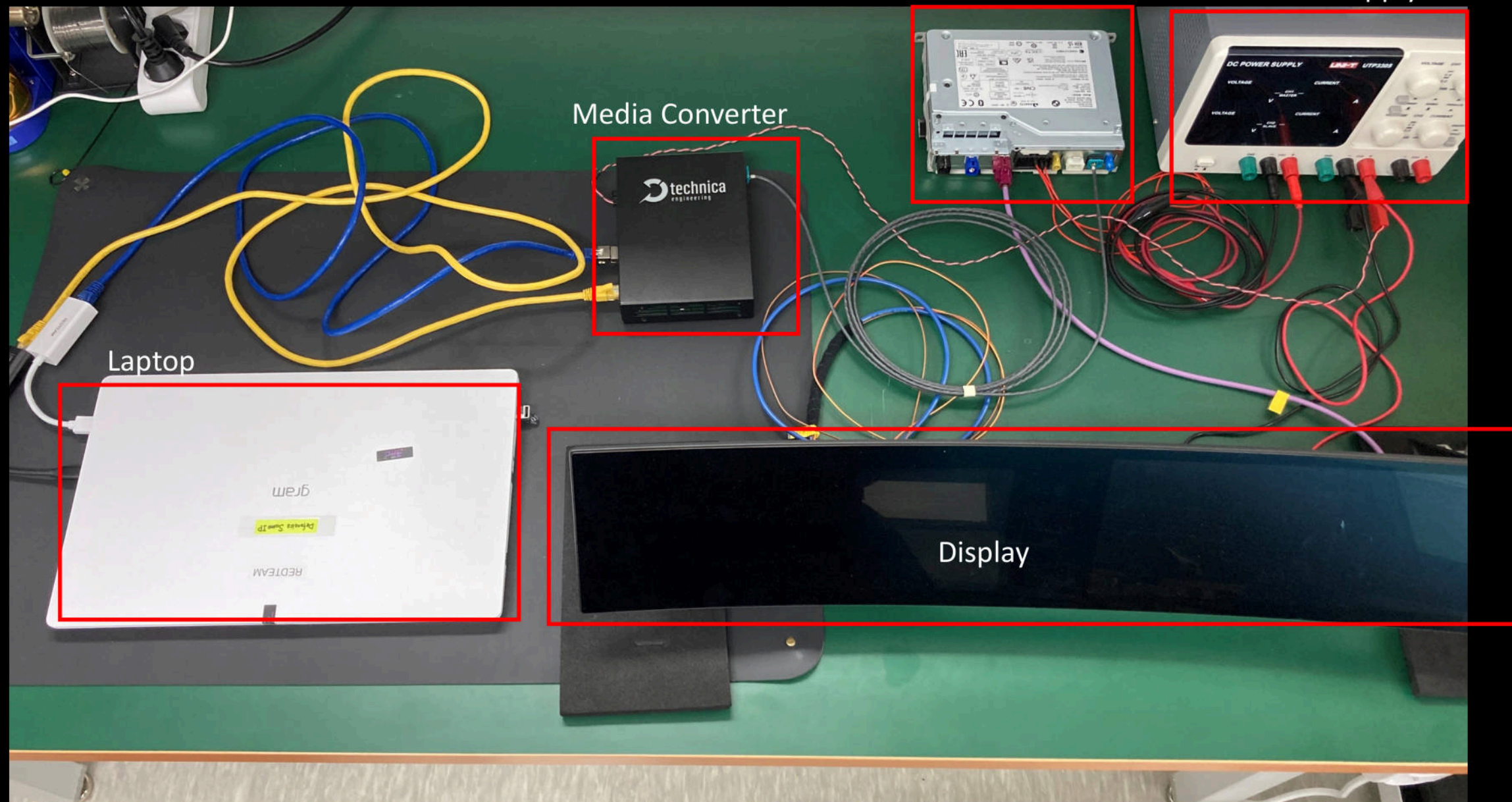
2017 - 2019 TESLA MODEL 3 CAR COMPUTER MEDIA CONTROLLE...
Pre-Owned
\$919.95 Was: \$699.96 8% off Buy It Now
+\$69.21 shipping estimate from United States
3 watchers

Setup Test Environment

Overview



Conversion between Automotive Ethernet and standard Gigabit Ethernet



Laptop

Media Converter

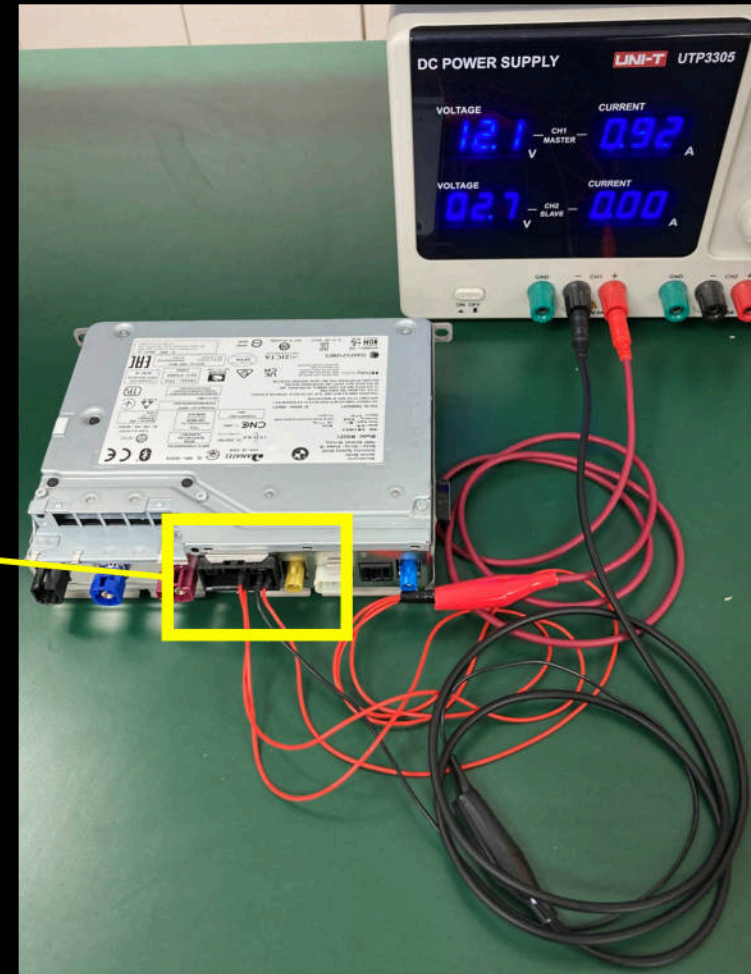
ECU

Power Supply

Display

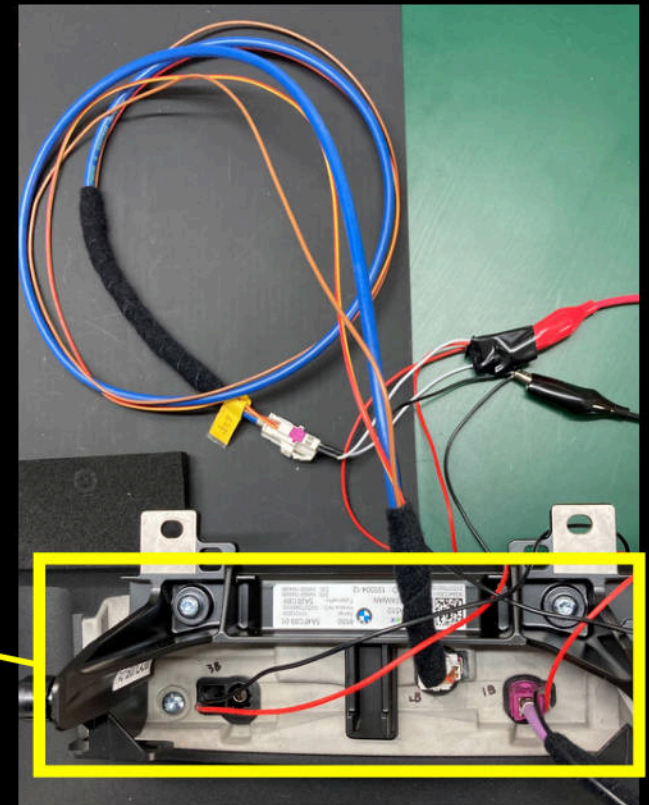
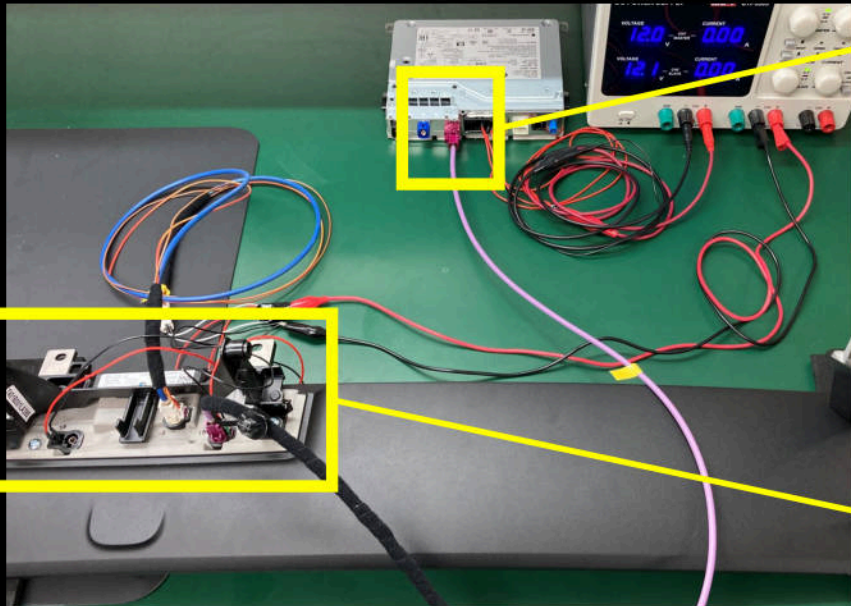
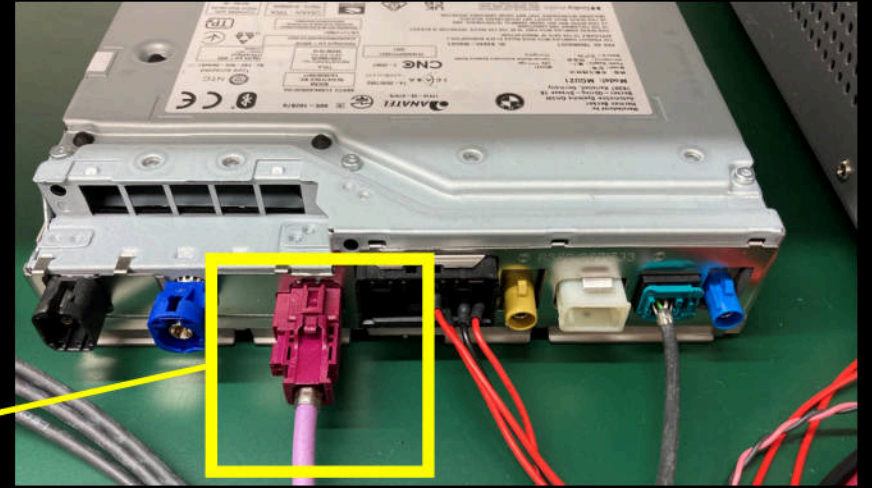
Connect power to ECU

- Red (+)
 - Power (12V)
 - Ethernet wake up (12V)
- Black (-)
 - Ground

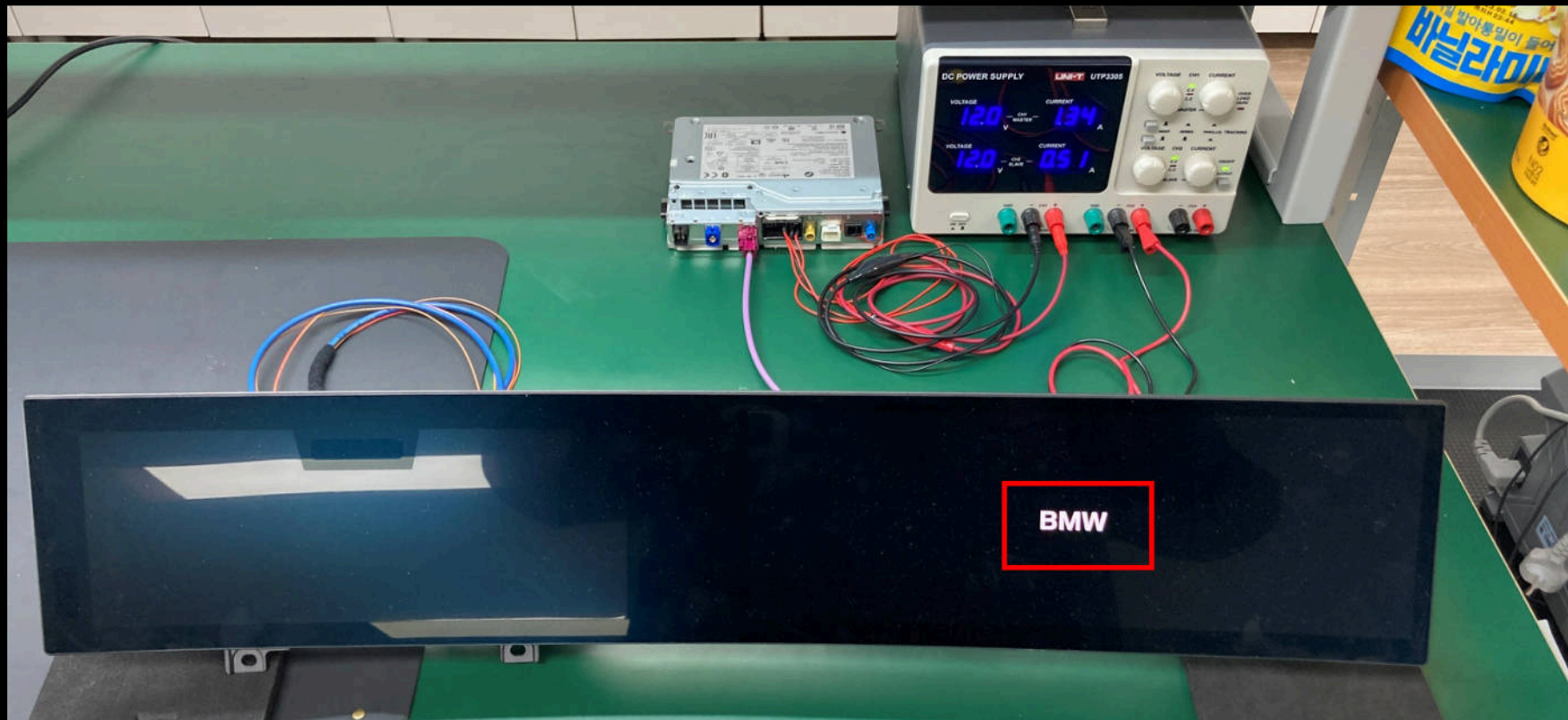


You can check the pin map from TIS.

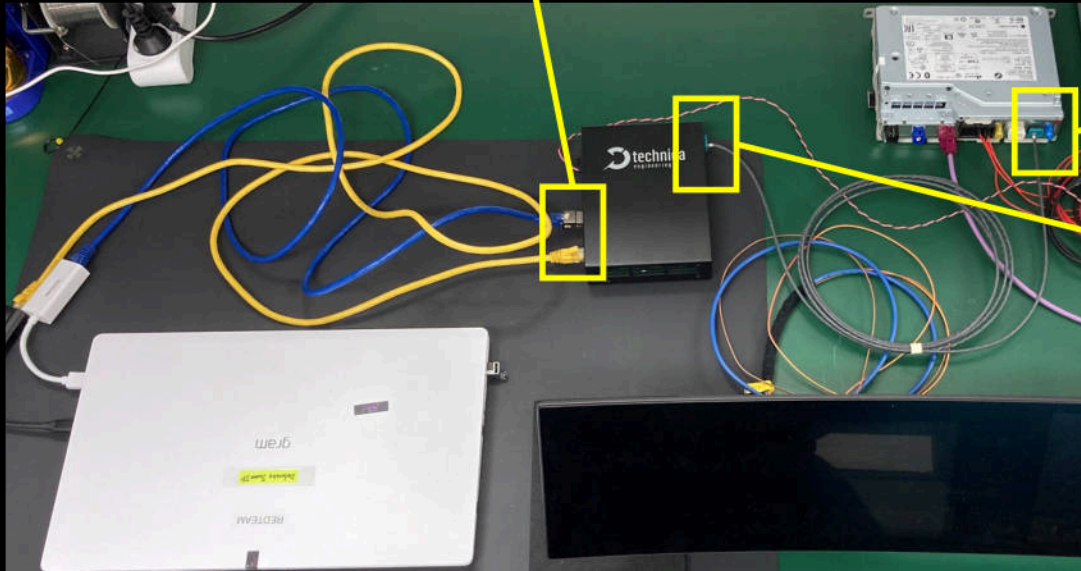
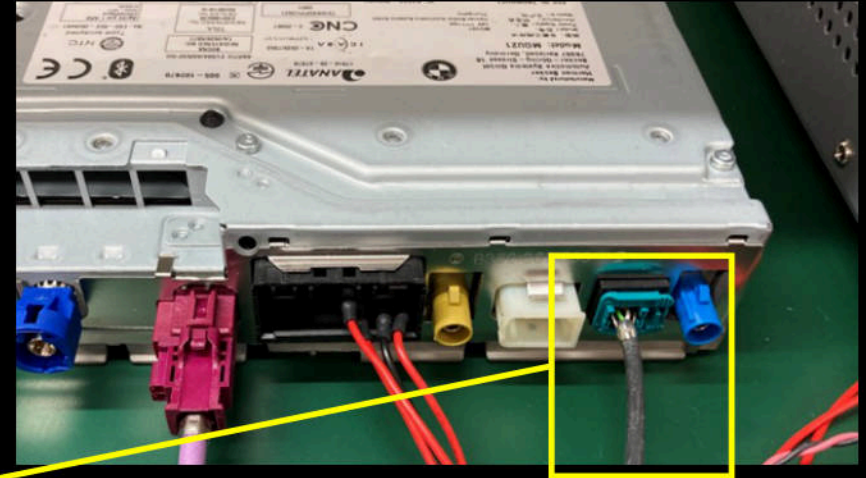
Setup display



Power On!



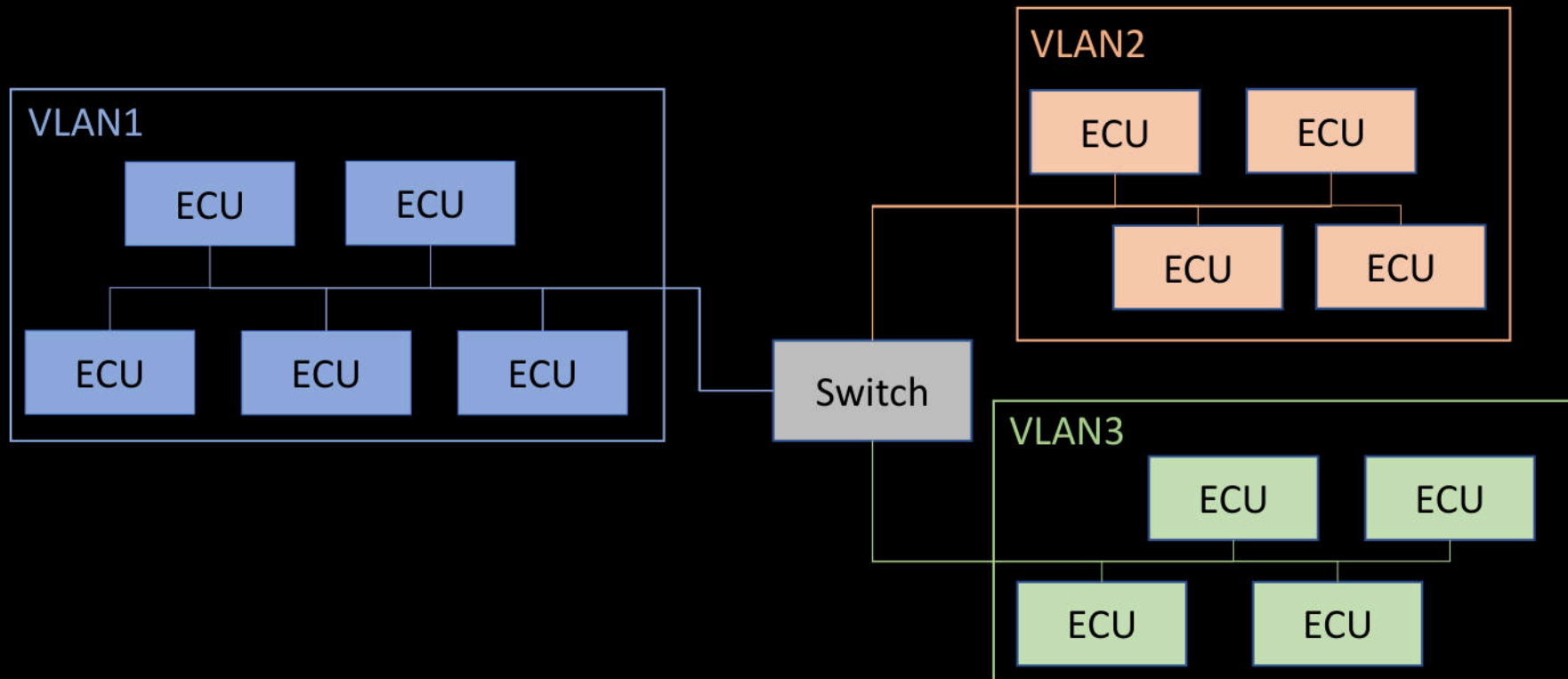
ECU – Media Converter – Laptop



Network Configuration for SOME/IP

Virtual LAN (VLAN) for Automotive Ethernet

- Automotive Ethernet uses VLAN to separate in-vehicle network logically.

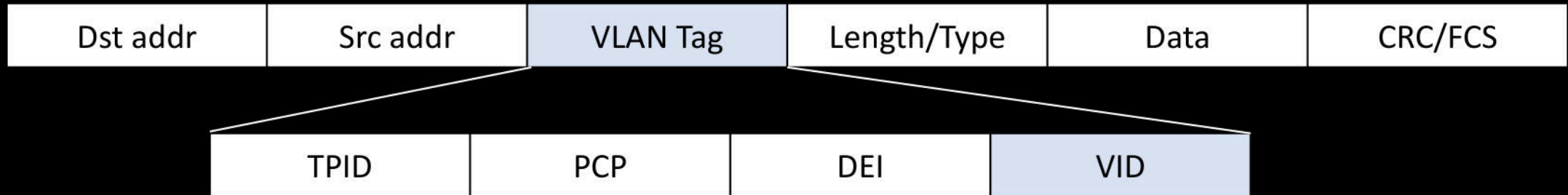


Virtual LAN (VLAN) for Automotive Ethernet

Traditional Ethernet frame



Ethernet frame with VLAN tag



How to know IP and VLAN ID?

Wireshark shows IP and VLAN ID.

(Some USB to Ethernet adapter doesn't show VLAN ID. If you can't see VLAN ID, it will be a good idea to change the adapter.)

No.	Time	Source	Destination	Protocol	Length	Info
85	4.64...	160.48.199.99	239.192.255.251	SOME/IP-SD	470	SOME/IP Service Discovery Protocol [Offer]
86	4.64...	160.48.199.99	239.192.255.251	SOME/IP-SD	346	SOME/IP Service Discovery Protocol [Find]
87	4.88...	160.48.199.99	239.192.255.251	SOME/IP-SD	470	SOME/IP Service Discovery Protocol [Offer]
88	4.92...	160.48.199.99	239.192.255.251	SOME/IP-SD	598	SOME/IP Service Discovery Protocol [Offer]
89	4.93...	160.48.199.99	239.192.255.251	SOME/IP-SD	762	SOME/IP Service Discovery Protocol [Offer]
90	4.93...	160.48.199.99	239.192.255.251	SOME/IP-SD	106	SOME/IP Service Discovery Protocol [Find]
91	4.96...	160.48.199.99	239.192.255.251	SOME/IP-SD	762	SOME/IP Service Discovery Protocol [Offer]
92	4.96...	160.48.199.99	239.192.255.251	SOME/IP-SD	106	SOME/IP Service Discovery Protocol [Find]
93	5.02...	160.48.199.99	239.192.255.251	SOME/IP-SD	762	SOME/IP Service Discovery Protocol [Offer]
94	5.02...	160.48.199.99	239.192.255.251	SOME/IP-SD	106	SOME/IP Service Discovery Protocol [Find]
95	5.14...	160.48.199.99	239.192.255.251	SOME/IP-SD	106	SOME/IP Service Discovery Protocol [Find]
96	5.14...	160.48.199.99	239.192.255.251	SOME/IP-SD	762	SOME/IP Service Discovery Protocol [Offer]
97	5.38...	160.48.199.99	239.192.255.251	SOME/IP-SD	762	SOME/IP Service Discovery Protocol [Offer]
98	5.43...	160.48.199.99	239.192.255.251	SOME/IP-SD	330	SOME/IP Service Discovery Protocol [Find]
99	5.46...	160.48.199.99	239.192.255.251	SOME/IP-SD	330	SOME/IP Service Discovery Protocol [Find]
1	5.52...	160.48.199.99	239.192.255.251	SOME/IP-SD	330	SOME/IP Service Discovery Protocol [Find]

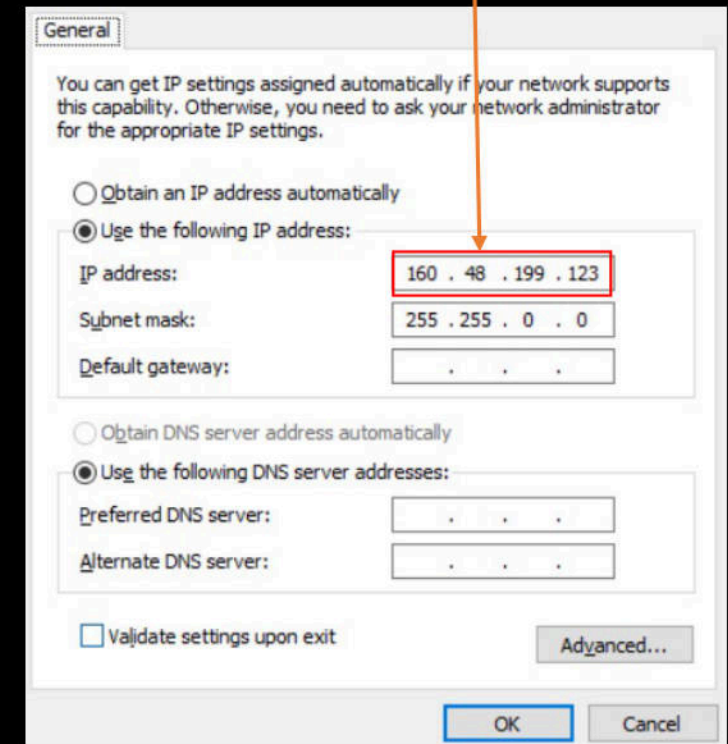
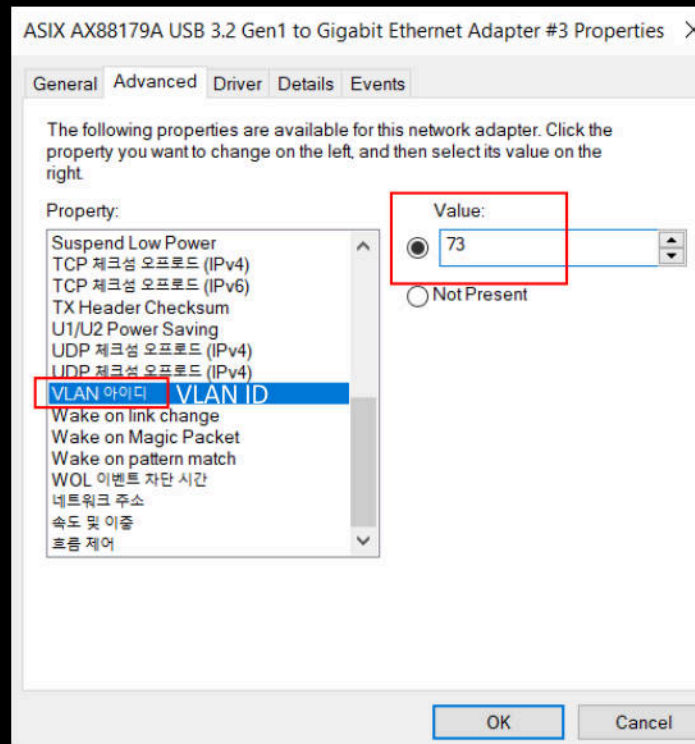
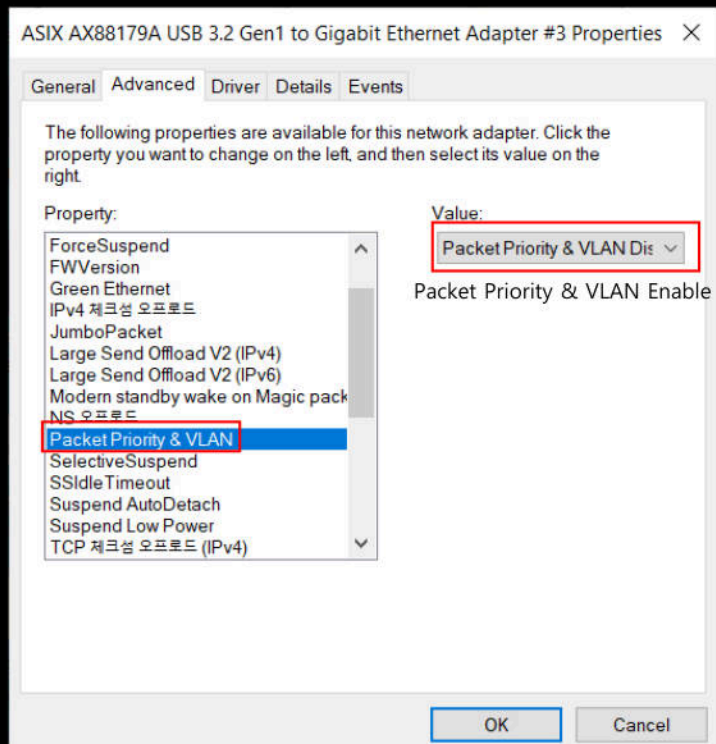
> Frame 91: 762 bytes on wire (6096 bits), 762 bytes captured (6096 bits) on interface \Device\NPF_{285000B2-E4}
> Ethernet II, Src: HarmanBe_23:04:1b (38:45:54:23:04:1b), Dst: IPv4mcast_40:ff:fb (01:00:5e:40:ff:fb)
> 802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 73
> Internet Protocol Version 4, Src: 160.48.199.99, Dst: 239.192.255.251
> User Datagram Protocol, Src Port: 30490, Dst Port: 30490
> SOME/IP Protocol (Service ID: 0xffff, Method ID: 0x8100, Length: 708)
> SOME/IP Service Discovery Protocol

Setup IP and VLAN

(Windows) Ethernet properties → Networking → Configure → Advanced

- "PacketPriority & VLAN" → "Packet Priority & VLAN Enable"
- "VLAN ID" → "Value" → *vlan id*

Select any IP in same subnetwork



Discovery SOME/IP Services

SOME/IP Service Discovery (SD)

Several SOME/IP services are running on the different ports at ECU.
To fuzz a SOME/IP service, we should know the port and the service ID.

- Check “OFFER” message
ECU periodically broadcast “OFFER” containing the service IDs and port numbers.

- Send “FIND” message
ECU will respond with “OFFER”.

“OFFER” message includes service ID and port

6031	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	98	SOME/IP Service Discovery Protocol [Offer]
6032	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	110	SOME/IP Service Discovery Protocol [Offer]
6033	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	110	SOME/IP Service Discovery Protocol [Offer]
6034	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	110	SOME/IP Service Discovery Protocol [Offer]
6035	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	98	SOME/IP Service Discovery Protocol [Offer]
6036	494.7...	160.48.199.99	160.48.199.123	SOME/IP-SD	98	SOME/IP Service Discovery Protocol [Offer]

Frame 6032: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface \Device\NPF_{285000B2-E4C4-4000-8000-000000000000}

Ethernet II, Src: HarmanBe_23:04:1b (38:45:54:23:04:1b), Dst: Celink_17:3d:fb (a0:ce:c8:17:3d:fb)

Internet Protocol Version 4, Src: 160.48.199.99, Dst: 160.48.199.123

User Datagram Protocol, Src Port: 30490, Dst Port: 30490

SOME/IP Protocol (Service ID: 0xffff, Method ID: 0x8100, Length: 60)

- Service ID: 0xffff
- Method ID: 0x8100
- Length: 60
- Client ID: 0x0000
- Session ID: 0x0006
- SOME/IP Version: 0x01
- Interface Version: 0x01
- Message Type: 0x02 (Notification)
- Return Code: 0x00 (Ok)

SOME/IP Service Discovery Protocol

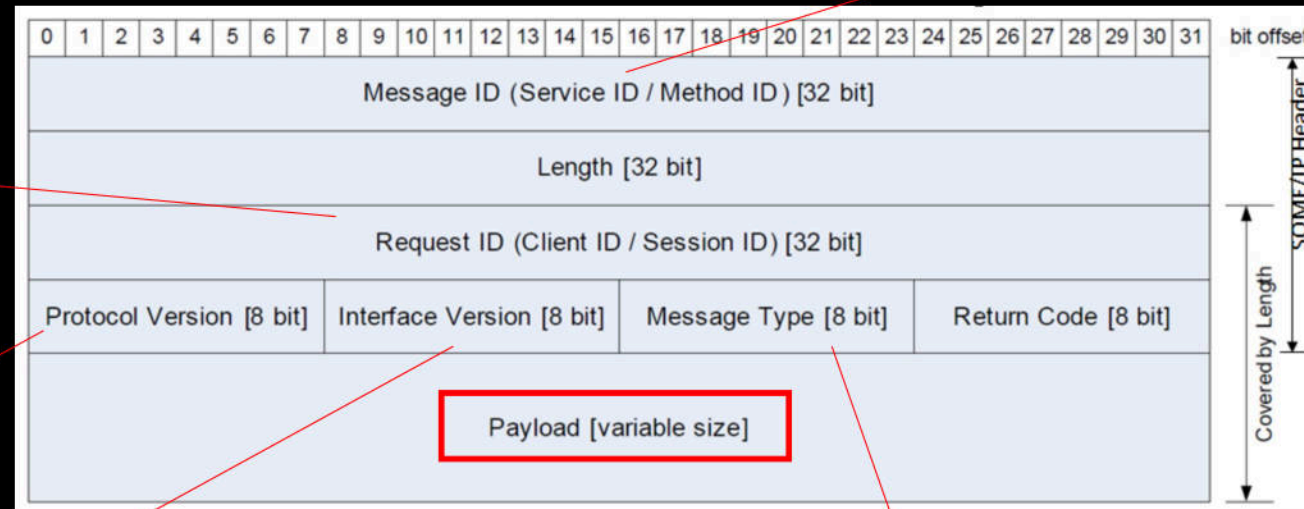
- Flags: 0xc0, Reboot Flag, Unicast Flag
- Reserved: 0x000000
- Length of Entries Array: 16
- Entries Array
 - Offer Service Entry (Service ID 0xb020, Instance ID 0x0001, Version 1.1)
- Length of Options Array: 24
- Options Array
 - 0: IPv4 Endpoint Option (160.48.199.99:30506 (TCP))
 - 1: IPv4 Endpoint Option (160.48.199.99:30506 (UDP))

Let's Do Fuzzing!

Fuzz Data Generation

Service ID [16 Bit]	0 [1 Bit]	Method ID [last 15 Bit]
---------------------	-----------	-------------------------

Client ID [16 Bits]	Session ID [16 Bits]
---------------------	----------------------



0x01

0x01

Number	Value	Description
0x00	REQUEST	A request expecting a response (even void)
0x01	REQUEST_NO_RETURN	A fire&forget request
0x02	NOTIFICATION	A request of a notification/event callback expecting no response
0x80	RESPONSE	The response message
0x81	ERROR	The response containing an error)
0x20	TP_REQUEST	A TP request expecting a response (even void)
0x21	TP_REQUEST_NO_RETURN	A TP fire&forget request
0x22	TP_NOTIFICATION	A TP request of a notification/event callback expecting no response
0x23	TP_RESPONSE	The TP response message
0x24	TP_ERROR	The TP response containing an error)

How to know ECU's state?

- Return codes
 - We can guess the ECU's state and error handling logics.
- Check response on the valid SOME/IP-SD message
 - no response from ECU → SOME/IP service is dead?
- Check periodic SOME/IP-SD messages from the ECU
 - no longer “*OFFER*” messages transmission from the ECU → SOME/IP service is dead?

Return Codes	Name
0x00	E_OK
0x01	E_NOT_OK
...	...
0x06	E_TIMEOUT
0x09	E_MALFORMED_MESSAGE

Implementation: Scapy supports SOME/IP!

LAYER-SPECIFIC DOCUMENTATION

- Layers
 - Automotive-specific Documentation**
 - Overview
 - Technical Background
 - Layers
 - CAN
 - CANSockets
 - CAN Calibration Protocol (CCP)
 - Universal calibration and measurement protocol (XCP)
 - ISOTP
 - UDS
 - GMLAN
 - Ecu Utility examples
 - SOME/IP and SOME/IP SD messages**
 - OBD
 - Test-Setup Tutorials

SOME/IP and SOME/IP SD messages

Creating a SOME/IP message

This example shows a SOME/IP message which requests a service 0x1234 with the method 0x421. Different types of SOME/IP messages follow the same procedure and their specifications can be seen here http://www.some-ip.com/papers/cache/AUTOSAR_TR_SomeIpExample_4.2.1.pdf.

Load the contribution:

```
load_contrib('automotive.someip')
```

Create UDP package:

```
u = UDP(sport=30509, dport=30509)
```

Create IP package:

```
i = IP(src="192.168.0.13", dst="192.168.0.10")
```

Create SOME/IP package:

```
sip = SOMEIP()  
sip.iface_ver = 0  
sip.proto_ver = 1  
sip.msg_type = "REQUEST"  
sip.retcode = "E_OK"  
sip.srv_id = 0x1234  
sip.method_id = 0x421
```

You can insert fuzz data here
Add the payload:

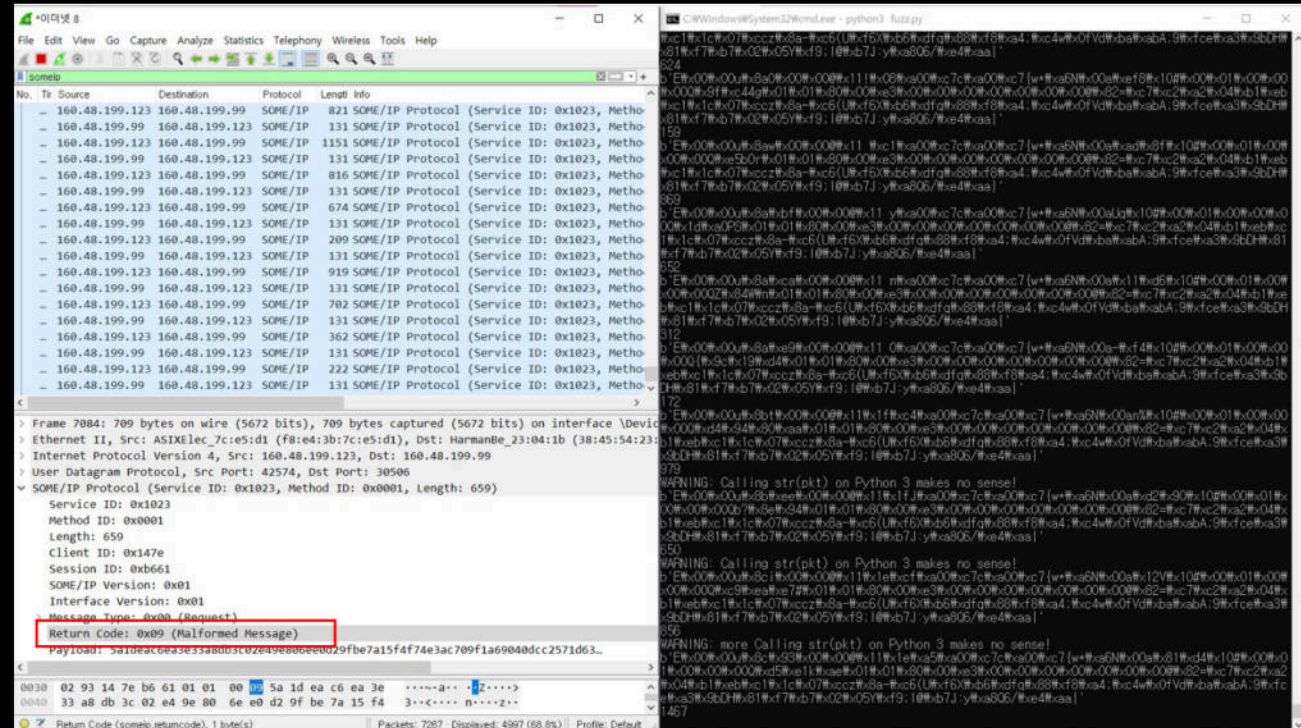
```
sip.add_payload(Raw("Hello"))
```

Stack it and send it:

```
p = i/u/sip  
send(p)
```

Results

- BMW
 - Still, we do fuzzing ...
 - Haven't found anything yet



- But we found some crashes of other OEM's ECUs.
 - we can't disclose it because of NDA.

Conclusion

- You can do car hacking without a whole car itself.
 - You can buy, setup and test ECUs.
- Let's do fuzzing Automotive Ethernet.
 - It's an early stage. There's still much to fuzz.
 - We will try DoIP.